Tinghui Zhu

Email | Website | Google Scholar

Research Interests

Self-evolving language and multimodal agents; long-horizon reasoning; learning from interaction and verifiable feedback.

Education

University of California, Davis 2025/09 – present
Ph.D. in Computer Science. Advisor: Muhao Chen Davis, CA

Fudan University 2022/09 – 2025/06
M.S. in Computer Science. Advisor: Yanghua Xiao Shanghai, China

Fudan University 2018/09 – 2022/06
B.S. in Computer Science. Shanghai, China

Experience

Google DeepMind 2026/06 – present
Software Engineering Intern, *Mentor: Wenxuan Zhou* Mountain View, CA

- Developed an online memory-management loop for long-horizon agents using counterfactual credit assignment to determine how memories should be updated; implemented a Git-native store that makes memory changes attributable and reversible.
- Evaluated the system across five model families and three long-horizon agent benchmarks; active memory management mitigated regressions from unmanaged memory and improved agent performance by up to 17.8%.

University of California, Davis, LUKA Group 2024/04 – 2025/06
Research Intern, *Mentor: Muhao Chen*

- Studied cross-modality parametric knowledge conflicts in large vision-language models, developing a pipeline for conflict detection, interpretation, and inference-time mitigation.
- Investigated modality extension and model merging for omni-modal models, analyzing their effects on language capability, cross-modal transfer, and generalization.

The Ohio State University, OSU NLP Group 2023/07 – 2024/02
Research Intern, *Mentor: Yu Su*

- Proposed Deductive Beam Search, integrating beam search with deductive reasoning to reduce error accumulation in LLMs. Developed a bootstrapping method for synthesizing erroneous reasoning steps and validated the approach across diverse model scales. [**COLM 2024**]
- Co-developed *TravelPlanner*, a tool-use planning benchmark with 1,225 curated planning intents and a sandbox over approximately four million records for evaluating real-world planning under multiple constraints. [**ICML 2024 Spotlight**]

Meituan, NLU group 2022/06 – 2023/03

Research Intern, *Mentor: Rui Xie*

- Developed a visual taxonomy expansion framework that integrates visual features into conceptual representations through prototype learning and cross-modal constraints, improving accuracy by 8.75% on the Chinese taxonomy benchmark. [**ACMMM 2023**]
- Built an automated taxonomy expansion pipeline that predicts hypernym relations between queries and clicked items from user interaction data.

Projects

Early Experience Scaling 2026/01 – 2026/06

- Built and released a 696K-example training corpus spanning eight embodied, tool-use, planning, and web-agent environments, supporting imitation learning, self-reflection, and implicit world modeling.
- Developed reproducible fine-tuning and evaluation infrastructure; reproduced consistent improvements over imitation learning across seven evaluated environments, with gains of up to 16.4 percentage points.

Publications

* Indicates equal contributions.

Published

1. **Tinghui Zhu**, Sheng Zhang, James Y Huang, Selena Song, Xiaofei Wen, Yuankai Li, Hoifung Poon, Muhao Chen. Video Models Can Reason with Verifiable Rewards. In *Proceedings of The Fortieth Annual Conference on Neural Information Processing Systems* (**NeurIPS**), 2026.
2. **Tinghui Zhu***, Kai Zhang*, Muhao Chen, Yu Su. Is Extending Modality The Right Path Towards Omni-Modality?. *ICLR 2026 Workshop on Multimodal Intelligence* (**ICLR Workshop**), 2026.
3. Sirui Xia, Aili Chen, Xintao Wang, **Tinghui Zhu**, Yikai Zhang, Jiangjie Chen, Yanghua Xiao. Can LLMs Learn to Map the World from Local Descriptions?. In *Proceedings of The 64th Annual Meeting of the Association for Computational Linguistics* (**ACL**), 2026.
4. Caiyu Hu, Yikai Zhang, **Tinghui Zhu**, Yiwei Ye, Yanghua Xiao. MCiteBench: A Multimodal Benchmark for Generating Text with Citations. In *Proceedings of the 30th Conference on Empirical Methods in Natural Language Processing* (**EMNLP**), 2025.
5. **Tinghui Zhu***, Kai Zhang*, Jian Xie, Yu Su. Deductive Beam Search: Decoding Deducible Rationale for Chain-of-Thought Reasoning. In *Proceedings of the 1st Conference on Language Modeling* (**COLM**), 2024.
6. Siye Wu*, Jian Xie*, Jiangjie Chen, **Tinghui Zhu**, Kai Zhang, Yanghua Xiao. How Easily do Irrelevant Inputs Skew the Responses of Large Language Models?. In *Proceedings of the 1st Conference on Language Modeling* (**COLM**), 2024.
7. Jiangjie Chen, Xintao Wang, Rui Xu, Siyu Yuan, Yikai Zhang, Wei Shi, Jian Xie, Shuang Li, Ruihan Yang, **Tinghui Zhu**, Aili Chen, Nianqi Li, Lida Chen, Caiyu Hu, Siye Wu, Scott Ren, Ziquan Fu, Yanghua Xiao. From Persona to Personalization: A Survey on Role-Playing Language Agents. Transactions on Machine Learning Research (**TMLR**), 2024.

8. Jian Xie*, Kai Zhang*, Jiangjie Chen, **Tinghui Zhu**, Renze Lou, Yuandong Tian, Yanghua Xiao, Yu Su. TravelPlanner: A Benchmark for Real-World Planning with Language Agents. In *Proceedings of the 42nd International Conference on Machine Learning* (**ICML Spotlight**), 2024.
9. **Tinghui Zhu**, Jingping Liu, Jiaqing Liang, Haiyun Jiang, Yanghua Xiao, Zongyu Wang, Rui Xie, Yunsen Xian. Towards Visual Taxonomy Expansion. In *Proceedings of the 32nd ACM Multimedia Conference* (**ACMMM**), 2023.
10. Chao Wang*, **Tinghui Zhu***, Zhixu Li, Jingping Liu. SLR: A Million-Scale Comprehensive Crossword Dataset for Simultaneous Learning and Reasoning. **Neurocomputing**, 2023.
11. Lihan Chen, **Tinghui Zhu**, Jingping Liu, Jiaqing Liang, Yanghua Xiao. End-to-end Entity Linking with Hierarchical Reinforcement Learning. In *Proceedings of the 37th Annual AAAI Conference on Artificial Intelligence* (**AAAI**), 2023.

Preprints

1. Xiaofei Wen, Wenjie Jacky Mo, Xingyu Fu, Rui Cai, **Tinghui Zhu**, Wendi Li, Yanan Xie, Muhao Chen, Peng Qi. When Vision Speaks for Sound. *Under Review*, 2026.
2. Yuankai Li, **Tinghui Zhu**, Ha Min Son, Zhe Zhao, Xin Liu, Muhao Chen. AQuaUI: Visual Token Reduction for GUI Agents with Adaptive Quadtrees. *Under Review*, 2026.
3. Yixu Huang, **Tinghui Zhu**, Muhao Chen. Learning Adaptive Reasoning Paths for Efficient Visual Reasoning. *Under Review*, 2026.
4. Boyu Zhu, Xiaofei Wen, Wenjie Jacky Mo, **Tinghui Zhu**, Yanan Xie, Peng Qi, Muhao Chen. OmniGuard: Unified Omni-Modal Guardrails with Deliberate Reasoning. *Under Review*, 2025.
5. James Y Huang, Sheng Zhang, Qianchu Liu, Guanghui Qin, **Tinghui Zhu**, Tristan Naumann, Muhao Chen, Hoifung Poon. Be My Eyes: Extending Large Language Models to New Modalities Through Multi-Agent Collaboration. *Under Review*, 2025.
6. **Tinghui Zhu**, Qin Liu, Fei Wang, Zhengzhong Tu, Muhao Chen. Unraveling Cross-Modality Knowledge Conflicts in Large Vision-Language Models. *Under Review*, 2024.

Awards

- **Graduate Fellowship**, University of California, Davis 2025/09

Services

Conference reviewer: ACL (2023–2026); COLM (2024–2026); EMNLP (2024–2026); ICLR (2025–2026); NeurIPS (2026).